

The Boltzmann equation for anisotropic metals

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It is pointed out that the Boltzmann equation in the theory of metals reduces to a finite set of linear algebraic equations whenever the scattering probability can be written as a finite sum of the form $\sum p_i(\mathbf{k}) q_i(\mathbf{k}')$. The explicit form of the solution, valid for arbitrary energy surfaces, is derived for this model.

1. INTRODUCTION

In the presence of an applied electric field $\mathbf{E}$ and a temperature gradient ∇T the electron distribution function in a metal is $f_0(\mathbf{k}) + f_1(\mathbf{k})$, where f_0 is the equilibrium distribution and where, when the scattering is elastic, f_1 satisfies the linear integral equation (Ziman 1960, p. 267)

$$\mathbf{u} \cdot \mathbf{v} = \int \{f_1(\mathbf{k}) - f_1(\mathbf{k}')\} \rho(\mathbf{k}, \mathbf{k}') dS' / \hbar v'. \quad (1)$$

Here
$$\mathbf{u} = -e\mathbf{E} \frac{\partial f_0}{\partial \epsilon} - \nabla T \frac{\partial f_0}{\partial T}, \quad (2)$$

and $\epsilon(\mathbf{k})$ is the energy, $\mathbf{k}$ the wave vector and $\mathbf{v} = \hbar^{-1} \partial \epsilon / \partial \mathbf{k}$ the velocity of an electron. The integral on the right-hand side of (1) refers to a single energy surface $\epsilon(\mathbf{k}) = \text{constant}$, and $\rho(\mathbf{k}, \mathbf{k}')$ is the scattering probability for the transition from the state $\mathbf{k}$ on the energy surface to the state $\mathbf{k}'$ in the region dS' of the energy surface where the magnitude of the velocity is v' . In metals one is primarily interested in the particular energy surface (the Fermi surface) which corresponds to the surface of the Fermi distribution for a degenerate electron gas.

In the special case in which the energy surfaces are spheres, and in which $\rho(\mathbf{k}, \mathbf{k}')$ depends only upon the angle between $\mathbf{k}$ and $\mathbf{k}'$, equation (1) possesses a well-known exact solution (Ziman 1960, p. 268), but little work has been done on the solution of (1) under more general conditions. Radcliffe (1955) has obtained an exact solution, valid for arbitrary energy surfaces, for the special case (of interest in the theory of semiconductors) in which $\rho(\mathbf{k}, \mathbf{k}')$ is constant (see also Herring 1955, appendix A); but the usual procedure in the theory of metals (Ziman 1960, *passim*) is to avoid any attempt to solve equation (1) directly and to base the discussion of the transport phenomena upon the equivalent variational problem, with the use of simple trial functions to represent the function $f_1(\mathbf{k})$. It is difficult to estimate the accuracy of the results obtained in this way, and recent work on the transport properties of the monovalent metals (Ziman 1961 *a, b*; Collins & Ziman 1961) has made it clear that more exact solutions of the Boltzmann equation are an essential prerequisite for a realistic theory of the transport phenomena in metals.

We shall here obtain the exact solution of equation (1) in closed form, for arbitrary energy surfaces, on the assumption that $\rho(\mathbf{k}, \mathbf{k}')$ can be written as a finite sum of products of a function of $\mathbf{k}$ only with a function of $\mathbf{k}'$ only, so that

$$\rho(\mathbf{k}, \mathbf{k}') = \sum_{i=1}^n p_i(\mathbf{k}) q_i(\mathbf{k}'), \quad (3)$$

where p_i, q_i are arbitrary functions of $\mathbf{k}$ of integrable square. The consequences of this model for the Boltzmann equation (1) do not seem to have been explored so far, although similar ideas occur, in a somewhat different context, in a paper by Herring & Vogt (1956, appendix A). When $\rho(\mathbf{k}, \mathbf{k}')$ is of the form (3) we have the well-known case in the theory of Fredholm equations where the kernel is of finite rank and the equation reduces to a finite system of linear algebraic equations (Smithies 1958, §3.2). In this case the problem of solving the Boltzmann equation is essentially trivial, and the transport phenomena can be calculated exactly from the explicit solution (equation (36) below) of the equation in terms of integrals of given functions over the Fermi surface. Since any general kernel can be replaced to an arbitrary degree of approximation by a kernel of finite rank of a sufficiently high order, the method given here can be regarded as an approximation method for the solution of equation (1) for arbitrary $\rho(\mathbf{k}, \mathbf{k}')$. When n is large, however, the method becomes cumbersome in practice and may not be the most advantageous for a direct numerical solution of the Boltzmann equation. But for models for which the number of terms in (3) is not too large (for example, when $\rho(\mathbf{k}, \mathbf{k}')$ is a polynomial of low degree in $|\mathbf{k} - \mathbf{k}'|^2$ or in $\cos \theta$, where θ is the angle between $\mathbf{k}$ and $\mathbf{k}'$) it should prove useful to have an elementary exact solution of the Boltzmann equation which can be used, for example, to test the accuracy of variational trial solutions.

2. SOLUTION OF THE BOLTZMANN EQUATION

2.1. We first simplify equation (1) slightly by introducing a vector distribution function $\mathbf{g}(\mathbf{k})$ defined by

$$f_1(\mathbf{k}) = \mathbf{u} \cdot \mathbf{g}(\mathbf{k}). \quad (4)$$

Then, since $\mathbf{u}$ is an arbitrary constant vector, $\mathbf{g}(\mathbf{k})$ satisfies the integral equation

$$\mathbf{v} = \int \{\mathbf{g}(\mathbf{k}) - \mathbf{g}(\mathbf{k}')\} \rho(\mathbf{k}, \mathbf{k}') dS' / \hbar v'. \quad (5)$$

(I am indebted to Mr P. Taylor for this remark.)

The homogeneous equation obtained by writing zero on the left-hand side of (5) has the non-zero solution $\mathbf{g}(\mathbf{k}) = \text{constant}$, and equation (5) is therefore insoluble unless the left-hand side is orthogonal to the solution of the transposed homogeneous equation (Smithies 1958, p. 50). In accordance with the principle of microscopic reversibility $\rho(\mathbf{k}, \mathbf{k}')$ must be such that

$$\rho(\mathbf{k}, \mathbf{k}') = \rho(\mathbf{k}', \mathbf{k}), \quad (6)$$

and the solution of the transposed homogeneous equation is therefore also $\mathbf{g}(\mathbf{k}) = \text{constant}$. The solubility condition (which is also obtained at once on integrating both sides of (5) over the energy surface and using (6)) is therefore

$$\int \mathbf{v} dS / \hbar v = 0, \quad (7)$$

and this is always satisfied since $\mathbf{v}$ is antisymmetric in the Brillouin zone. The solution of equation (5) is, however, not unique; the particular integral required in the physical applications is linear in $\mathbf{v}$ and reduces to zero when $\mathbf{v} = 0$.

2.2. To proceed with the solution when $\rho(\mathbf{k}, \mathbf{k}')$ has the form (3) it is convenient to regard the functions p_i, q_i ($i = 1, 2, \dots, n$) as column (ket) vectors in a real n -dimensional vector space and to write

$$|p\rangle = \{p_1, p_2, \dots, p_n\}, \quad |q\rangle = \{q_1, q_2, \dots, q_n\}. \quad (8)$$

The corresponding row (bra) vectors are denoted by $\langle p|$, $\langle q|$, and (3) becomes a scalar product

$$\rho(\mathbf{k}, \mathbf{k}') = \langle p(\mathbf{k}) | q(\mathbf{k}') \rangle = \langle p(\mathbf{k}') | q(\mathbf{k}) \rangle = \langle q(\mathbf{k}) | p(\mathbf{k}') \rangle \quad (9)$$

(where (6) has been used to obtain the second form).

With (9), equation (5) becomes

$$\mathbf{v} = \mathbf{g}(\mathbf{k}) \langle p(\mathbf{k}) | Q \rangle - \langle p(\mathbf{k}) | \mathbf{C} \rangle, \quad (10)$$

where $|Q\rangle = \int |q(\mathbf{k}')\rangle dS' / \hbar v'$ and $|\mathbf{C}\rangle = \int \mathbf{g}(\mathbf{k}') |q(\mathbf{k}')\rangle dS' / \hbar v'$ (11)

are n -vectors which are constant on the energy surface. (Note that $|\mathbf{C}\rangle$ is a vector both in $\mathbf{k}$ -space and in n -space.) Solving (10) for $\mathbf{g}(\mathbf{k})$ we have

$$\mathbf{g}(\mathbf{k}) = \{\mathbf{v} + \langle p(\mathbf{k}) | \mathbf{C} \rangle\} / \langle p(\mathbf{k}) | Q \rangle, \quad (12)$$

and substituting this back into the expression for $|\mathbf{C}\rangle$ we obtain

$$|\mathbf{C}\rangle = |\mathbf{W}\rangle + A |\mathbf{C}\rangle, \quad (13)$$

where $|\mathbf{W}\rangle = \int \frac{\mathbf{v}' |q(\mathbf{k}')\rangle}{\langle p(\mathbf{k}') | Q \rangle} \frac{dS'}{\hbar v'}$, $A = \int \frac{|q(\mathbf{k}')\rangle \langle p(\mathbf{k}')|}{\langle p(\mathbf{k}') | Q \rangle} \frac{dS'}{\hbar v'}$. (14)

A is a dyad in n -space, its ij -element being

$$\int \frac{q_i(\mathbf{k}') p_j(\mathbf{k}')}{\langle p(\mathbf{k}') | Q \rangle} \frac{dS'}{\hbar v'};$$

note that the matrix A is not, in general, symmetric. Writing (13) in the form

$$(I - A) |\mathbf{C}\rangle = |\mathbf{W}\rangle, \quad (15)$$

where I is the unit $n \times n$ matrix, we have obtained a system of n linear algebraic equations for the determination of the components of the unknown n -vector $|\mathbf{C}\rangle$.

2.3. The system (15) is similar in structure to the original integral equation (5) and possesses similar properties. We note first that

$$A |Q\rangle = \int \frac{|q(\mathbf{k}')\rangle \langle p(\mathbf{k}') | Q \rangle}{\langle p(\mathbf{k}') | Q \rangle} \frac{dS'}{\hbar v'} = |Q\rangle, \quad (16)$$

and also that, if $\tilde{A}$ is the transposed matrix associated with A and

$$|P\rangle = \int |p(\mathbf{k}')\rangle dS' / \hbar v', \quad (17)$$

$$\tilde{A} |P\rangle = \int \frac{|p(\mathbf{k}')\rangle \langle q(\mathbf{k}') | P \rangle}{\langle p(\mathbf{k}') | Q \rangle} \frac{dS'}{\hbar v'} = |P\rangle, \quad (18)$$

since $\langle q(\mathbf{k}')|P\rangle = \langle p(\mathbf{k}')|Q\rangle$ by virtue of (9). Thus $|Q\rangle$ is an eigenvector of A with eigenvalue unity, and $|P\rangle$ is an eigenvector of $\tilde{A}$ with eigenvalue unity. The matrix $I - A$ is therefore singular, and the system (15) is only soluble provided that the right-hand side is orthogonal to the solution $|X\rangle = |P\rangle$ of the transposed homogeneous system $(I - \tilde{A})|X\rangle = 0$ (and the solution is then not unique). Now, using (15) and (18), we have

$$\begin{aligned}\langle P|W\rangle &= \langle P|I - A|C\rangle = \langle P|C\rangle - \langle P|A|C\rangle \\ &= \langle P|C\rangle - \langle C|\tilde{A}|P\rangle \\ &= \langle P|C\rangle - \langle C|P\rangle \\ &= 0,\end{aligned}\tag{19}$$

so that (15) is always soluble whenever (5) is soluble.

We now assume that the eigenvalue 1 of A occurs only once, so that the rank of the matrix $I - A$ is $n - 1$. (This is physically equivalent to the assumption that the energy is the only 'constant of the motion' for electrons making transitions from one point on the energy surface to another.) Under these conditions the solution of the system (15) follows standard lines (Ferrar 1941, chap. 8) and can be obtained explicitly as follows.

We introduce as a basis for the n -space the linearly independent unit vectors $|e_1\rangle, |e_2\rangle, \dots, |e_n\rangle$, where $|e_j\rangle$ is the vector whose j th component is 1, the other components being zero. Then, for any arbitrary n -vector $|X\rangle$ with components $X_1, X_2, \dots, X_n$, we have

$$|X\rangle = |e_1\rangle X_1 + |e_2\rangle X_2 + \dots + |e_n\rangle X_n.\tag{20}$$

Now normalize $|Q\rangle$ so that $Q_1 = 1$ (this can always be done, by suitable renumbering of the components of $|Q\rangle$ if necessary, since at least one of the components of $|Q\rangle$ must be non-zero), and let H be the non-singular matrix

$$H = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ Q_2 & 1 & 0 & \dots & 0 \\ Q_3 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ Q_n & 0 & 0 & \dots & 1 \end{pmatrix},\tag{21}$$

with an inverse

$$H^{-1} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ -Q_2 & 1 & 0 & \dots & 0 \\ -Q_3 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ -Q_n & 0 & 0 & \dots & 1 \end{pmatrix},\tag{22}$$

which is such that

$$H^{-1}|Q\rangle = |e_1\rangle.\tag{23}$$

(The operators $I - H, I - H^{-1}$ are projection operators on to the $(n - 1)$ -dimensional subspace generated by the unit vectors $|e_2\rangle, |e_3\rangle, \dots, |e_n\rangle$.)

Now (15) is equivalent to

$$H^{-1}(I-A)HH^{-1}|C\rangle = (I-B)H^{-1}|C\rangle = H^{-1}|W\rangle, \quad (24)$$

where

$$B = H^{-1}AH, \quad (25)$$

and, since $(I-A)|Q\rangle = 0$, the equation $(I-B)|X\rangle = 0$ has the solution

$$|X\rangle = H^{-1}|Q\rangle = |e_1\rangle. \quad (26)$$

The general solution of (24) is therefore

$$H^{-1}|C\rangle = \lambda|e_1\rangle + |R\rangle, \quad (27)$$

where λ is an arbitrary vector in $\mathbf{k}$ -space which is constant on the energy surface, and where $|R\rangle$ satisfies

$$(I-B)|R\rangle = H^{-1}|W\rangle \quad (28)$$

and is such that

$$\langle e_1|R\rangle = 0. \quad (29)$$

By explicit matrix multiplication we find that

$$H^{-1}|W\rangle = |W\rangle - W_1\{|Q\rangle - |e_1\rangle\} = |Z\rangle, \quad \text{say,} \quad (30)$$

and that B is the appropriately partitioned matrix

$$B = \begin{pmatrix} 1 & A_{12} & A_{13} & \dots & A_{1n} \\ 0 & B_{22} & B_{23} & \dots & B_{2n} \\ 0 & B_{32} & B_{33} & \dots & B_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & B_{n2} & B_{n3} & \dots & B_{nn} \end{pmatrix}, \quad (31)$$

where

$$B_{ij} = \begin{vmatrix} 1 & A_{1j} \\ Q_i & A_{ij} \end{vmatrix} \quad (i, j = 2, 3, \dots, n). \quad (32)$$

If $|X'\rangle$ denotes the projection of any n -vector $|X\rangle$ on to the subspace generated by $|e_2\rangle, \dots, |e_n\rangle$, we may now determine $|R'\rangle$ by replacing (28) by the $n-1$ equations

$$(I-B')|R'\rangle = |Z'\rangle, \quad (33)$$

where B' is the $(n-1) \times (n-1)$ submatrix, with elements given by (32), obtained from B by omitting the first row and column, and where I is now the unit $(n-1) \times (n-1)$ matrix. (The components of $|Z'\rangle$ are given by $Z_i = W_i - W_1 Q_i$, $i = 2, 3, \dots, n$.) The matrix $I - B'$ is non-singular, and the solution of (33) is

$$|R'\rangle = (I - B')^{-1}|Z'\rangle, \quad (34)$$

where $(I - B')^{-1}$ is the matrix inverse to $I - B'$.

This completes the solution. The vector $|R\rangle$ is completely defined by (29) and (34), and (27) (with (21)) gives

$$\begin{aligned} |C\rangle &= \lambda H|e_1\rangle + H|R\rangle \\ &= \lambda|Q\rangle + |R\rangle. \end{aligned} \quad (35)$$

Substituting this into (12), we have finally

$$g(\mathbf{k}) = \lambda + \{v + \langle p(\mathbf{k})|R\rangle\} / \langle p(\mathbf{k})|Q\rangle. \quad (36)$$

For the physical applications we take $\lambda = 0$ without loss of generality.

3. DISCUSSION

3.1. With the distribution function given in the form (36) the determination of the transport magnitudes such as the electrical conductivity or the thermoelectric power is a standard formal calculation (see Ziman 1960, chap. 7) which will not be carried out here, since the results are of interest primarily in connexion with applications to particular models. The detailed investigation of such applications lies beyond the scope of the present paper, and we confine ourselves to some general comments on the theory and to an explicit evaluation of (36) for one simple illustrative example.

3.2. $|p\rangle$ and $|q\rangle$ appear unsymmetrically in (36), but this asymmetry is only apparent. In fact the symmetry relation (9) shows that an equivalent solution exists in which $|p\rangle$ and $|q\rangle$ are interchanged. From this it follows easily that the solution satisfies all the usual symmetry requirements of transport theory which arise as a consequence of the principle of microscopic reversibility. It is also easy to verify the invariance of (36) for orthogonal transformations which leave the scalar products (9) invariant.

3.3. The (anisotropic) time of relaxation $\tau(\mathbf{k})$ is defined as $\mathbf{u} \cdot \dot{\mathbf{g}}(\mathbf{k})/(\mathbf{u} \cdot \mathbf{v})$ (Ziman 1960, p. 505) and will clearly depend on the direction of $\mathbf{u}$ (and hence of the applied fields) unless the $\mathbf{k}$ -space vector $\langle p(\mathbf{k}) | \mathbf{R} \rangle$ is parallel to $\mathbf{v}$. It is only when this condition is satisfied that $\tau(\mathbf{k})$ may be regarded as a truly scalar quantity; in the general case, when $\dot{\mathbf{g}}$ and $\mathbf{v}$ are not parallel, the conduction process may be characterized by a tensor time of relaxation $T(\mathbf{k})$ which is such that $\dot{\mathbf{g}}(\mathbf{k}) = T(\mathbf{k}) \mathbf{v}$.

3.4. In practice it will often be possible to simplify the general result (36) considerably as a consequence of crystal symmetry requirements. In particular, a drastic simplification occurs whenever the matrix A is diagonal. In this case, writing $A_{ij} = A_i \delta_{ij}$, we see from (32) that $B_{ij} = A_{ij} = A_i \delta_{ij}$ for $i, j = 2, 3, \dots, n$, so that $(I - B')^{-1} = (1 - A_i)^{-1} \delta_{ij}$ ($i, j = 2, 3, \dots, n$). The numbers A_i are the eigenvalues of A and, with the eigenvalue $A_1 = 1$ (corresponding to the eigenvector $|Q\rangle$) non-degenerate, we must have $Q_2 = Q_3 = \dots = Q_n = 0$, i.e. (with the same normalization as before) $|Q\rangle = |e_1\rangle$. Hence $\langle p(\mathbf{k}) | Q \rangle = p_1(\mathbf{k})$, $|Z\rangle = |W\rangle$, and the components of $|\mathbf{R}\rangle$ are $\mathbf{R}_1 = 0$, $\mathbf{R}_i = \mathbf{W}_i/(1 - A_i)$ ($i = 2, 3, \dots, n$). Equation (36) (with $\lambda = 0$) therefore reduces to the much simpler expression

$$\dot{\mathbf{g}}(\mathbf{k}) = \frac{1}{p_1(\mathbf{k})} \left\{ \mathbf{v} + \sum_{i=2}^n \frac{p_i(\mathbf{k}) \mathbf{W}_i}{1 - A_i} \right\}. \quad (37)$$

We note further that, in the special case where $\rho(\mathbf{k}, \mathbf{k}')$ is a single product $p(\mathbf{k})q(\mathbf{k}')$, $|\mathbf{R}\rangle = 0$ and the solution is simply

$$\dot{\mathbf{g}}(\mathbf{k}) = \mathbf{v}/p(\mathbf{k}). \quad (38)$$

The result given by Radcliffe (1955) for the case $\rho(\mathbf{k}, \mathbf{k}') = \text{constant}$ is a special case of this.

3.5. As a simple though somewhat unphysical illustration of the present formalism we shall evaluate the time of relaxation numerically for the case where the Fermi surface consists of the curved surface of a circular cylinder of radius k_0 and

length $2b$, and where $\rho(\mathbf{k}, \mathbf{k}') = \rho_0 + \rho_1 \cos \theta$, θ being the angle between $\mathbf{k}$ and $\mathbf{k}'$. We choose the z axis along the axis of the cylinder and $\mathbf{u}$ normal to this axis, say along the x -direction, so that we require the x -component $g_x(\mathbf{k})$ of $\mathbf{g}(\mathbf{k})$. For this model

$$\epsilon(\mathbf{k}) = \alpha(k_x^2 + k_y^2) = \alpha k_0^2, \quad \mathbf{v} = (2\alpha/\hbar)(k_x, k_y, 0), \quad v' = (2\alpha/\hbar)k_0, \quad (39)$$

where α is a constant, and all the integrals over the Fermi surface are easily evaluated by introducing cylindrical coordinates in the $\mathbf{k}$ -space. We take (with appropriate normalization)

$$|p(\mathbf{k})\rangle = \beta \left\{ \rho_0, \rho_1 \frac{k_x}{|\mathbf{k}|}, \rho_1 \frac{k_y}{|\mathbf{k}|}, \rho_1 \frac{k_z}{|\mathbf{k}|} \right\}, \quad |q(\mathbf{k})\rangle = \frac{1}{\beta} \left\{ 1, \frac{k_x}{|\mathbf{k}|}, \frac{k_y}{|\mathbf{k}|}, \frac{k_z}{|\mathbf{k}|} \right\}, \quad (40)$$

where $\beta = 2\pi b/\alpha$, and obtain from (11) and (14)

$$|Q\rangle = \{1, 0, 0, 0\}, \quad (41)$$

$$|W_x\rangle = \left\{ 0, \frac{\alpha^2 k_0^2}{2\pi \hbar b^2 \rho_0} \sinh^{-1} \frac{b}{k_0}, 0, 0 \right\}, \quad (42)$$

and $A_{ij} = A_i \delta_{ij}$, with

$$A_1 = 1, \quad A_2 = A_3 = \frac{k_0 \rho_1}{2b \rho_0} \tan^{-1} \frac{b}{k_0}, \quad A_4 = \frac{k_0 \rho_1}{b \rho_0} \left(\frac{b}{k_0} - \tan^{-1} \frac{b}{k_0} \right). \quad (43)$$

Since A is diagonal, we may substitute into (37) to obtain

$$g_x(\mathbf{k}) = \frac{\alpha v_x}{2\pi b \rho_0} \left\{ 1 + \frac{\frac{k_0^2 \rho_1}{2b \rho_0 \sqrt{(k_0^2 + k_z^2)}} \sinh^{-1} \frac{b}{k_0}}{1 - \frac{k_0 \rho_1}{2b \rho_0} \tan^{-1} \frac{b}{k_0}} \right\}. \quad (44)$$

The time of relaxation $\tau(\mathbf{k})$ is g_x/v_x , and, putting $\mu = b/k_0$, $\nu = \rho_1/\rho_0$, $\delta = k_z/k_0$ (where $0 \leq |\delta| \leq \mu$), we may write the result in the convenient form

$$\frac{\tau(\delta)}{\tau(0)} = \frac{1 + \frac{G(\mu, \nu)}{\sqrt{(1 + \delta^2)}}}{1 + G(\mu, \nu)}, \quad \text{with} \quad G(\mu, \nu) = \frac{(\nu/2\mu) \sinh^{-1} \mu}{1 - (\nu/2\mu) \tan^{-1} \mu}, \quad (45)$$

which gives the variation of τ with height along the cylinder.

It is interesting to compare this result with that obtained for the present model from the expression

$$\frac{1}{\tau(\mathbf{k})} = \int (1 - \cos \theta) \rho(\mathbf{k}, \mathbf{k}') \frac{dS'}{\hbar v'}, \quad (46)$$

which is exact in the standard isotropic case and which was used by Ziman (1961*b*) to estimate the anisotropy of $\tau(\mathbf{k})$ for the noble metals. With (46) we easily obtain, instead of (45),

$$\frac{\tau(\delta)}{\tau(0)} = \left\{ 1 - \frac{\delta^2}{1 + \delta^2} G'(\mu, \nu) \right\}^{-1}, \quad \text{with} \quad G'(\mu, \nu) = \frac{\nu \left(1 - \frac{3}{2} \frac{\tan^{-1} \mu}{\mu} \right)}{1 - (\nu/2\mu) \tan^{-1} \mu}. \quad (47)$$

The anisotropy of τ predicted by (47) may be quite different from that given by (45); in particular, for large μ (narrow cylinders) it is much too large and in the wrong sense. This is illustrated in figure 1, which shows the variation with μ of the maximum anisotropy ratio $\tau(\mu)/\tau(0)$ as predicted by the two formulae for $\nu = \frac{3}{4}$. (Note

that, as $\mu \rightarrow \infty$, $\tau(\mu)/\tau(0) \rightarrow 1$ according to (45), but $\tau(\mu)/\tau(0) \rightarrow 1/(1-\nu)$ according to (47). (46) grossly overemphasizes the effect of large-angle collisions in this limit.) It does not follow, of course, that the use of (46) must necessarily lead to large errors when more realistic and less extremely anisotropic models are considered, but it is clear that considerable caution is needed in the quantitative use of approximate expressions for the time of relaxation.*

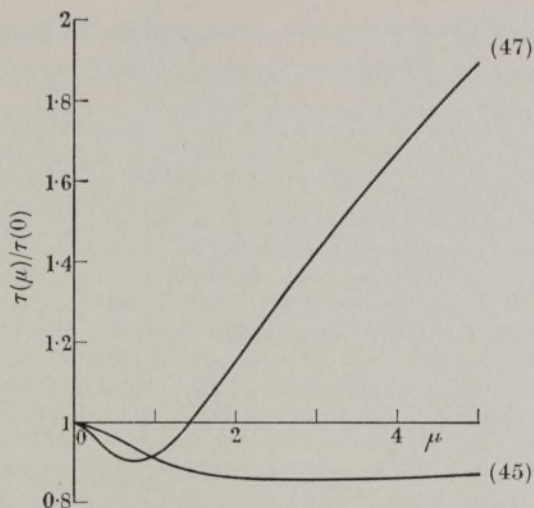


FIGURE 1. The anisotropy of the relaxation time for a cylindrical Fermi surface when $\rho(\mathbf{k}, \mathbf{k}')$ varies as $1 + \frac{1}{2} \cos \theta$, where θ is the angle between $\mathbf{k}$ and $\mathbf{k}'$. $\tau(\mu)$ and $\tau(0)$ are the values of τ at the ends and at the centre of the cylinder, and μ is the ratio length/diameter of the cylinder. The applied field is normal to the axis of the cylinder.

3.6. Finally, some ways will be indicated in which the present formalism can be generalized. The most important extension is to the case when a magnetic field is present, in view of the sensitive dependence of the galvanomagnetic effects upon the structure of the Fermi surface and the anisotropy of the relaxation time (Pippard 1960). In the presence of a magnetic field $\mathbf{H}$ there is an additional term

$$-(e/\hbar c)(\mathbf{v} \times \mathbf{H}) \cdot \partial f_1 / \partial \mathbf{k}$$

on the left-hand side of the Boltzmann equation (1), and, when $\rho(\mathbf{k}, \mathbf{k}')$ has the form (3), the methods of the present paper can immediately be applied to solve the equation as a series in ascending powers of $\mathbf{H}$; it is then necessary to solve an equation of the type of equation (1) at each stage of the process of successive approximations. The low-field galvanomagnetic effects, therefore, are readily evaluated, but the effects in strong fields present a problem of a greater order of mathematical difficulty. We note lastly that the discussion need not be restricted to the case of elastic scattering of electrons only. An inspection of the form of the generalized collision integrals which replace the right-hand side of equation (1) when more complicated

* [Note added in proof, 24 April 1962.] Dr Ziman has reminded me of his remark (Ziman 1961*a*) that (46) is likely to give a better approximation to τ if θ is taken to be the angle between the velocity vectors $\mathbf{v}$ and $\mathbf{v}'$ instead of the angle between $\mathbf{k}$ and $\mathbf{k}'$, and in fact most of the discrepancy between (45) and (46) disappears when (46) is re-interpreted in this way.

scattering processes are envisaged (for example, inelastic collisions in which both electrons and phonons take part; compare Ziman 1960, pp. 277, 357) shows that the Boltzmann equation is soluble in finite terms whenever the relevant scattering probability can be written as an appropriate generalization of equation (3). But the extent to which the formalism will prove useful in practice in such complicated cases remains to be investigated.

I am deeply indebted to Mr P. Taylor for helping me to detect a serious error in the first version of the theory. I am also grateful to various colleagues, particularly Mr B. Fishel, for helpful discussions of the work.

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